

Supplemental Online Content

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This supplemental material has been provided by the authors to give readers additional information about their work.

Supplemental Online Content

Transcriptomic profiles in nasal epithelium and asthma endotypes in youth

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eMETHODS

Study populations

STAR

The Stress and Treatment Response in Puerto Rican and African American Children with Asthma Study (STAR) was a six-week study to examine chronic stress and response to inhaled corticosteroids (ICS) in youth with asthma. At both the Puerto Rico and Pittsburgh sites, the main recruitment tool was a screening questionnaire given to parents of youth with asthma. Subjects were eligible if they were 8 to 20 years old, had ≥ 3 Puerto Rican grandparents per parental report (in Puerto Rico) or ≥ 3 African American grandparents per parental report (in Pittsburgh), were steroid naïve (no nasal, inhaled, or systemic corticosteroids) for ≥ 4 weeks, and had: 1) physician-diagnosed mild to moderate persistent asthma (defined as symptoms >2 times/week *or* use of inhaled short-acting β_2 -agonists >2 times/week *or* ≥ 2 exacerbations requiring systemic corticosteroids in the previous year) *and* 2) bronchodilator responsiveness (BDR) $\geq 8\%^1$ *or*, if BDR $<8\%$, airway hyper-responsiveness to methacholine ($PC_{20} < 16$ mg/ml). Exclusion criteria included chronic diseases other than asthma, severe asthma (defined as any intubation for asthma after age 4 years *or* ≥ 3 hospitalizations *or* ≥ 6 visits to the emergency department (ED)/urgent care in the previous year *or* chronic use of medications other than ICS and/or leukotriene inhibitors and short-acting β_2 -agonists), current smoking or former smoking of ≥ 5 pack-years, an $FEV_1 < 60\%$ of predicted, and (for those recruited after March 2020), SARS-CoV-2 infection in the prior two weeks. After nasal specimen collection and education on inhaler use, participants were treated with daily medium-dose inhaled mometasone (100 mcg, 2 puffs once per day for children aged 8-11 years, and 200 mcg, 2 puffs once per day for children

12 years and older) for six weeks, with adherence monitored using a dose counter and emphasized at in-person visits (at 2 and 6 weeks) and at weekly phone calls between visits.

EVA-PR

Subject recruitment and study procedures for the Epigenetic Variation and Childhood Asthma in Puerto Ricans Study (EVA-PR) were previously reported in detail². EVA-PR was a case-control study of asthma and indicators of asthma severity or control in Puerto Ricans aged 9 to 20 years who were recruited from households in the metropolitan areas of San Juan and Caguas (PR), employing a multistage probability sampling design. Primary sampling units were randomly selected neighborhood clusters based on the 2000 U.S. census, and secondary units were randomly selected households within each primary sampling unit. A household was eligible if at least one resident was a person aged 9-20 years. Of the 1,111 selected households, 1,045 were contacted. Of these, 720 had at least 1 resident who met age and other inclusion criteria (four Puerto Rican grandparents [per parental report] and residence in the same household for at least 1 year). Participants with asthma (defined as physician-diagnosed asthma and at least one episode of wheeze in the previous year) were selected at random among households that had at least one resident who met criteria (one person was selected at random if there was more than one eligible person). Similarly, one person without asthma (no physician-diagnosed asthma or wheeze in the previous year) was randomly selected from the remaining households. To reach a target sample of ~500 youth, the study attempted to enroll a random sample (n=638) of these 720 households. Parents of 95 (14.9%) of these 638 households refused to participate or could not be reached, leaving 543 participants with (n=269) and without (n=274) asthma who consented and completed study interviews. Of the 269 participants with asthma, 237 had

complete data on nasal transcriptomics, blood biomarkers, and lung function measures and were thus included in this analysis.

VDKA

Subject recruitment and the design of the Vitamin D Kids Asthma Study (VDKA) were previously described in detail³. In brief, VDKA was a 48-week randomized, double-blinded, placebo-controlled trial of vitamin D₃ (4,000 IU/day) to prevent severe disease exacerbations in children with mild persistent asthma³. Children were recruited from 7 sites across the US, including the University of Pittsburgh Medical Center Children's Hospital of Pittsburgh (Pittsburgh, PA). Eligible participants were children with asthma, aged 6 to 16 years, with serum vitamin D levels <30 ng/mL but ≥ 10 ng/mL (until July 21, 2017) or ≥ 14 ng/mL (after that date, following a protocol amendment). Entry criteria included **i.** physician-diagnosed asthma for at least 1 year; **ii.** at least 1 severe asthma exacerbation (defined as use of systemic corticosteroids for at least 3 days, or a hospitalization or ED visit requiring systemic corticosteroids) in the previous year⁴; **iii.** use of asthma medications (daily controller medications, or inhaled β_2 -agonists at least thrice per week) for at least 6 months in the previous year; **iv.** forced expiratory volume in one second (FEV₁) $\geq 70\%$ of predicted; and **v.** either bronchodilator responsiveness $\geq 8\%$ or, in those without bronchodilator responsiveness, increased airway responsiveness to methacholine (a PC₂₀ <8 mg/ml if not receiving inhaled corticosteroids (ICS) or <6 mg/ml if receiving ICS). Exclusion criteria included chronic respiratory disorders other than asthma, chronic oral corticosteroid therapy, severe asthma, and inability to perform adequate spirometry. Before randomization, all participants had to complete a four-week run-in period. To be eligible for randomization, the participants had to have no severe asthma exacerbations and demonstrate

$\geq 75\%$ adherence with both low-dose inhaled Fluticasone (88 mcg twice per day in children aged 6-11 years and 110 mcg twice per day in children ≥ 12 years) and placebo capsules by the end of the run-in period. Nasal samples were collected from participants at the Pittsburgh site at visit 3 (the randomization visit), before randomization occurred. The ancillary study of nasal omics was approved by the IRB of the University of Pittsburgh.

Study protocols

In all studies (STAR, EVA-PR, and VDKA), similar questionnaires on demographics and respiratory health were administered by trained interviewers, and spirometry was conducted with an EasyOne spirometer (NDD Medical Technologies, Andover, Mass) according to American Thoracic Society and European Respiratory Society recommendations⁵. In STAR, FeNO was measured using a NIOX VERO instrument (Circassia AB, Morrisville, NC) following manufacturer protocols and expressed as parts per billion (ppb). In all studies, eosinophil count was measured in peripheral blood using Coulter-Counter techniques. Serum total IgE and IgE specific to common allergens (Der p 1 and Bla g 2 in all studies, plus Fel d 1, Can f 1 and Mus m 1 in EVA-PR) were measured using the Unicap 100 system (Pharmacia & Upjohn, Kalamazoo, MI).

RNA sequencing

For at least four weeks before nasal sampling, no participant in any study received systemic corticosteroids. Furthermore, all participants in STAR and most participants in EVA-PR (~98%) and VDKA (~94%) did not use nasal steroids for ≥ 4 weeks prior to nasal sampling. In all studies, RNA-Seq was conducted at the Genomics Core of the University of Pittsburgh using

RNA from nasal specimens collected from the inferior turbinate, as previously described². Library preparation was done using TruSeq Stranded Total RNA Library Prep Kit with Ribo-Zero Gold High Throughput kit in EVA-PR and TruSeq Stranded mRNA in STAR and VDKA, which remove both cytoplasmic and mitochondrial ribosomal RNA (rRNA) (Illumina, San Diego, CA), according to the manufacturer's protocol. Libraries were run on the Illumina NextSeq 500 platform in EVA-PR and on the Illumina NovaSeq 6000 S4-200 platform in STAR and VDKA, with paired-end 75-100 cycles and with 80-100 million reads per sample.

Samples were processed using the nf-core pipeline⁶. RNA-Seq quality control was conducted using FastQC⁷ and summarized by MultiQC. 3' adapters and bad-quality reads were trimmed with Trim Galore!⁸ and Cutadapt⁹. Using the STAR RNA-Seq aligner¹⁰, trimmed reads were aligned to the latest human reference genome at the time of the studies and annotated using RSEM¹¹. Samples with poor overall quality (outliers in PCA plots) and low alignment rate (<80%) were excluded from the analysis.

In all studies and before RNA extraction, CD326(+) nasal epithelial cells were selected from a subset of nasal samples for sensitivity analyses accounting for potential effects of different cell types. The current analysis was not adjusted for cell sorting because only eight markers were used, and principal component analysis plots in each of the three studies (**eFigure 4**) demonstrated no clear clustering by cell sorting protocol.

Statistical Analysis

We utilized the following R packages in our analyses: tidyverse 2.0.0, cluster 2.1.6, stats 4.4.0, ClusterR 1.3.2, DESeq2 1.44.0, WGCNA 1.72.5, clusterProfiler 4.12.0, org.Hs.eg.db 3.19.1, gplots 3.1.3.1, rpart 4.1.23, caret 6.0.94, NbClust 3.0.1, randomForest 4.7.1.1, pROC 1.18.5, ROCR 1.0.11, and plotROC 2.3.1.

Cluster Validation

eFigure 2 illustrates our approach to cluster validation. Despite identifying the eight signature genes, there is no clear threshold to directly classify patients into T2-high or T17-high categories. Gene expression levels can be affected by batch effects, platforms and sequencing depth variations across different cohorts; therefore, while T2-high patients are expected to have higher expression values for the T2 genes compared to T2-low patients, the actual classification necessitates an unsupervised clustering method to ensure patient classification.

We first used the eight signature genes to classify patients into three clusters using an unsupervised clustering method. We then labeled them according to three endotypes ($T2^{HIGH}/T17^{LOW}$, $T2^{LOW}/T17^{HIGH}$, and $T2^{LOW}/T17^{LOW}$) based the relative expression value of T2 3 gene mean and T17 5 gene mean. This step was essential to categorize the patients based on gene expression patterns; otherwise, we would be unable to define the three endotypes. To assess the robustness of our clustering, we examined the Average Proportion of Non-overlap (APN) to assess cluster stability.

After clustering, we compared the three groups' clinical characteristics and lung function measures to explore and study the characteristics of the clusters. To further validate that the T2^{HIGH} and the T17^{HIGH} clusters indeed capture T2-high and T17-high biological features, we conducted differential gene expression (DE) analysis using R package DESeq2 1.44.0 and meta-analysis and weighted gene co-expression network analysis (WGCNA) using R package WGCNA 1.72.5.

Calculations of predicted FeNO in EVA-PR and VDKA

We used elastic-net prediction models to estimate FeNO in EVA-PR and VDKA, since FeNO was not measured in these two studies. Models were first constructed in STAR, as it had measured FeNO, by splitting the STAR dataset into training (80%) and testing (20%) sets to assess prediction accuracy utilizing a set of 1,013 pre-selected genes. Candidate genes were identified through a differential expression (DE) analysis of FeNO using DESeq2¹² models adjusted for age, sex, household income, batch effects, cell sorting, and study site. Next, for prediction in the EVA-PR and VDKA datasets, all samples from STAR were used to build the model that was subsequently applied to the EVA-PR and VDKA datasets, with hyperparameters determined by the 5-Fold Cross-Validation.

Prediction models for T2^{HIGH} cluster assignment

To help clinicians develop an easy-to-use empirical decision rule, optimal cutoff points for FeNO, total IgE, and eosinophils to predict the T2^{HIGH} cluster in isolation (i.e., individually) were determined using Youden's Index¹³. Separately, a decision tree machine learning method¹⁴ was implemented to determine the optimal combination of FeNO, total IgE, and/or eosinophils,

and optimal cutoff points when these biomarkers were used in combination, to predict the T2^{HIGH} profile. Hence, the cutoff points determined for the biomarkers individually using Youden's Index and in combination using a decision tree machine learning method were different.

While simple decision trees provide clear interpretability, which is crucial for clinical applications, using more complex models such as Random Forest improves prediction accuracy. Because of this, we also conducted additional experiments using a Random Forest model. We performed two experiments: one with the full dataset and one with only the consensus data. In each scenario, we trained the model on the STAR cohort and evaluated its performance both internally (using 10-fold cross-validation) and externally (using the EVA-PR and VDKA cohorts).

Differential gene expression (DE) analysis and meta-analysis

A differential expression analysis was conducted using a multivariable negative binomial regression framework in DESeq2 to assess differentially expressed genes (DEGs) based on the transcriptomic endotype profile of consensus samples within each study. In EVA-PR, models were adjusted for sequencing batches, cell sorting, age, gender, income level, and the top 5 principal components generated from the genetics profile. In STAR, models were adjusted for study site, sequencing batches, cell sorting, age, gender, and income level. In VDKA, models were adjusted for age, sex, race, treatment arm, parental education, and cell sorting.

Subsequently, an inverse-variance weighted meta-analysis was performed among the three studies. DEGs were defined as FDR P-value less than 0.01. Up-regulated DEGs with meta log₂ fold change > 1 were selected for the pathway enrichment analysis.

Weighted gene co-expression network analysis (WGCNA)

We conducted a consensus Weighted Gene Co-expression Network Analysis (WGCNA) on three cohorts (STAR, EVAPR, and VDKA) to investigate the gene modules associated with asthma endotypes. First, we performed soft-thresholding power network topology analysis separately for each cohort and then determined a suitable power value that was consistent across all cohorts. We then constructed a network based on the adjacency matrix, which was then transformed into a topological overlap matrix (TOM) to estimate the distance between each pair of genes. Next, we employed hierarchical clustering with average and dynamic methods to construct a cluster tree and classify genes into different modules. To assess the associations between modules and asthma endotypes, we calculated the correlations of the consensus module eigengene (ME) with the endotypes for each cohort. Only correlations with the same direction among all three cohorts were retained, and the smallest correlation among cohorts was used to represent each gene module-phenotype correlation. Candidate gene modules were defined as those with correlations with endotypes greater than 0.3. Subsequently, we performed gene ontology (GO) functional enrichment analysis for modules related to T2 and T17. Significantly enriched GO terms were collected and visualized using bar plots.

Refining gene expression profiles for T2^{HIGH} and T17^{HIGH} asthma

Based on the gene modules we identified using WGCNA and genes that were significant (FDR $p < 0.01$) in the DE analysis, we built a new signature gene panel. Using the T2 and T17 candidate genes identified, we built Lasso binary classification prediction models for T2-high and T17-high asthma respectively using individuals in all three cohorts with a consensus profile.

eRESULTS

Our findings cannot be explained by seasonality or allergic rhinitis. Seasonality was not linked to transcriptomic profiles in our two largest studies, and self-reported naso-ocular symptoms apart from colds in the prior year (which are independent of healthcare access) did not vary across profiles in any cohort, while allergic rhinitis was more common in T2^{HIGH} profile than in T17^{HIGH} profile in STAR only.

Performance of the clustering analysis

Cluster silhouette plots for each clustering method across STAR, VDKA, and EVA-PR are shown in **eFigure 3**. **eTable 1** shows the consistency of cluster assignments across the various methods used. The adjusted Rand index, which measures the similarity between different clustering results, for each cohort are provided in **eTable 2**, and the Average Proportion of Non-overlap (APN) for each cohort is shown in **eTable 3**.

Characteristics associated with ICS responsiveness across transcriptomic profiles in STAR

We next examined characteristics associated with ICS responsiveness across transcriptomic profiles in STAR (**eTable 6**), the only study that measured this outcome. Median ICS adherence during the 6-week study was 94.7% (interquartile range=82.1-100%), with similar proportions of ICS responders across transcriptomic profiles (35-42%). A reduced FeNO after ICS treatment was associated with ICS responsiveness in the T2^{HIGH} profile, while female sex and lower eosinophils and FEV₁/FVC were associated with ICS responsiveness in the T17^{HIGH} profile. In the T2^{LOW}/T17^{LOW} profile, ICS responders were more likely to have lower FEV₁ and FEV₁/FVC but higher T17 5GM than non-responders.

T2^{HIGH} classification rule

A classification rule based on a decision tree machine learning method using combined biomarkers for all STAR participants showed that total IgE ≥ 418 IU/mL and FeNO ≥ 28 ppb accurately identified the T2^{HIGH} profile in 89%, 77%, and 78% of participants in STAR, EVA-PR, and VDKA, respectively (**eTable 8, eFigure 7**). When using the same decision tree learning method for only STAR participants with a consensus profile, the T2^{HIGH} profile was best identified by total IgE ≥ 418 IU/mL and eosinophils ≥ 428 cells/ μ L or FeNO ≥ 32 ppb, but accuracies were similar (**eTable 8, eFigure 7**). Using this classification rule, accuracies of 90%, 79%, and 74% for identifying the T2^{HIGH} profile were achieved in individuals with a consensus profile in STAR, EVA-PR, and VDKA, respectively. For further validation, we used an alternative approach (Random Forest modeling) for predicting the T2^{HIGH} profile, obtaining comparable results (**eTable 8**).

Refining gene expression profiles for T2^{HIGH} and T17^{HIGH} asthma

Using the gene modules identified in WGCNA and genes that were significant in the DE analysis, we identified 265 candidate genes for T2-high asthma and 328 candidate genes for T17-high asthma. Using Lasso binary classification prediction models, we identified *AADAC*, *ADRA2A*, *CDH26*, *CISH*, *COL6A1*, *CST1*, *DPP4*, *FETUB*, *GCNT4*, *NTRK2*, *TMEM71*, and *TPSB2* as the refined T2 panel and *ADTRP*, *CASP10*, *CD3E*, *CXCL6*, *DUOX2*, *DUSP2*, *EPS8L3*, *MTNR1A*, *NCOA7*, *NFKBIZ*, *RND1*, *SP140*, *TNFAIP2*, *TREM1*, *VNN1*, and *ZC3H12A* as the refined T17 panel. Heat maps of expression levels in the STAR , EVA-PR and VDKA cohorts are shown in **eFigure 9**.

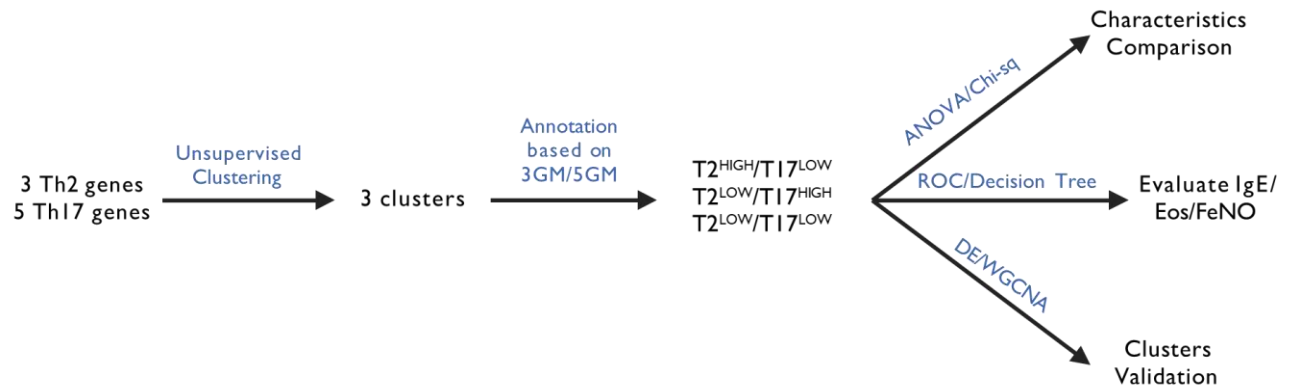
eFigure 1. Histograms of the T2 3-gene mean (top) and T17 5-gene mean (bottom) values in STAR, EVA-PR, and VDKA



Legend:

Abbreviations: STAR, Stress and Treatment Response in Puerto Rican and African American Children with Asthma Study; EVA-PR, Epigenetic Variation and Childhood Asthma in Puerto Ricans Study; VDKA, Vitamin D Kids Asthma Study.

eFigure 2. Approach for validation of clusters

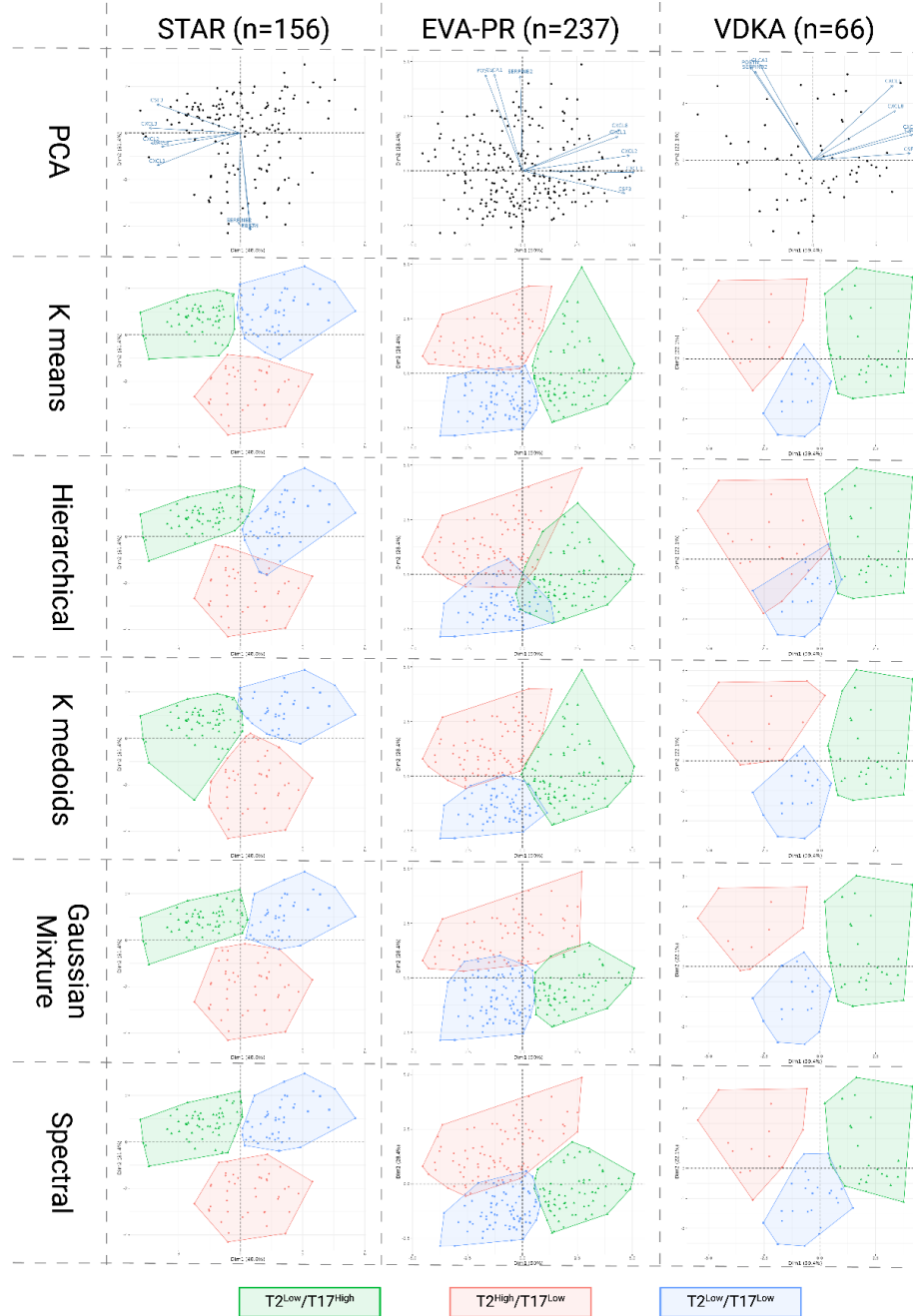


Legend:

Abbreviations: 3GM: 3 T2 genes mean; 5GM: 5 T17 genes mean; ANOVA, analysis of variance; ROC, receiver operator curve; IgE, Immunoglobulin E; Eos, eosinophils; FeNO, Fractional exhaled nitric oxide; DE, differential gene expression analysis; WGCNA, weighted gene co-expression network analysis.

Figure was created using BioRender.

eFigure 3. Cluster silhouette plots for each clustering method in STAR, EVA-PR, and VDKA

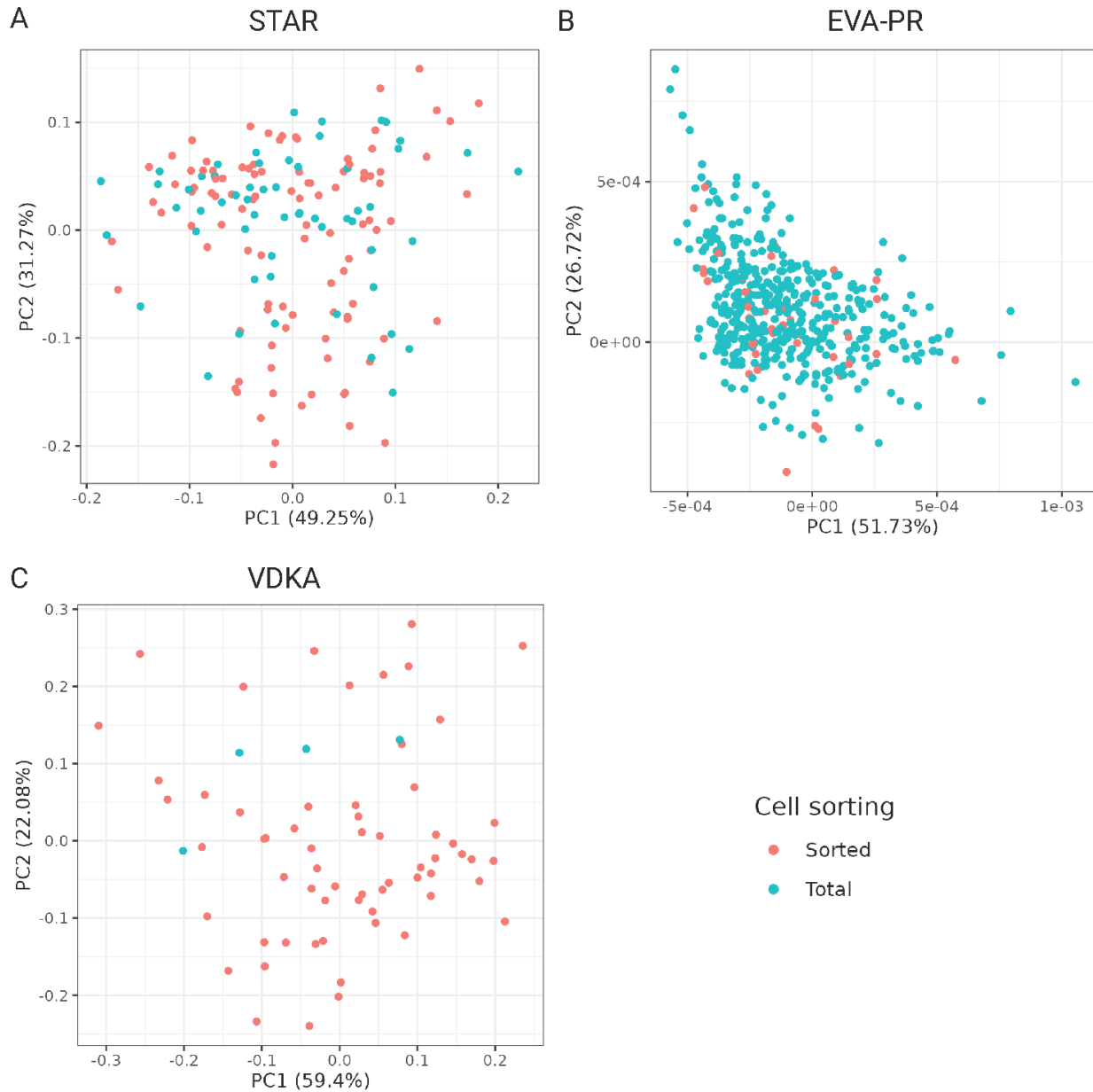


Legend:

Abbreviations: STAR, Stress and Treatment Response in Puerto Rican and African American Children with Asthma Study; EVA-PR, Epigenetic Variation and Childhood Asthma in Puerto Ricans Study; VDKA, Vitamin D Kids Asthma Study; PCA = Principal Components Analysis, which simplifies data by transforming original variables into new, uncorrelated principal components.

Dim 1 and Dim 2 represent the first and second principal components, showing the percentage of total variance each PC explains. Blue arrows indicate the contributions of the original variables, with direction showing influence and length representing the strength of the contribution. The silhouette illustrates the profile assignment of each clustering method. Figure was created using BioRender.

eFigure 4. Principal components analysis plots of cell sorting for STAR, EVA-PR, and VDKA

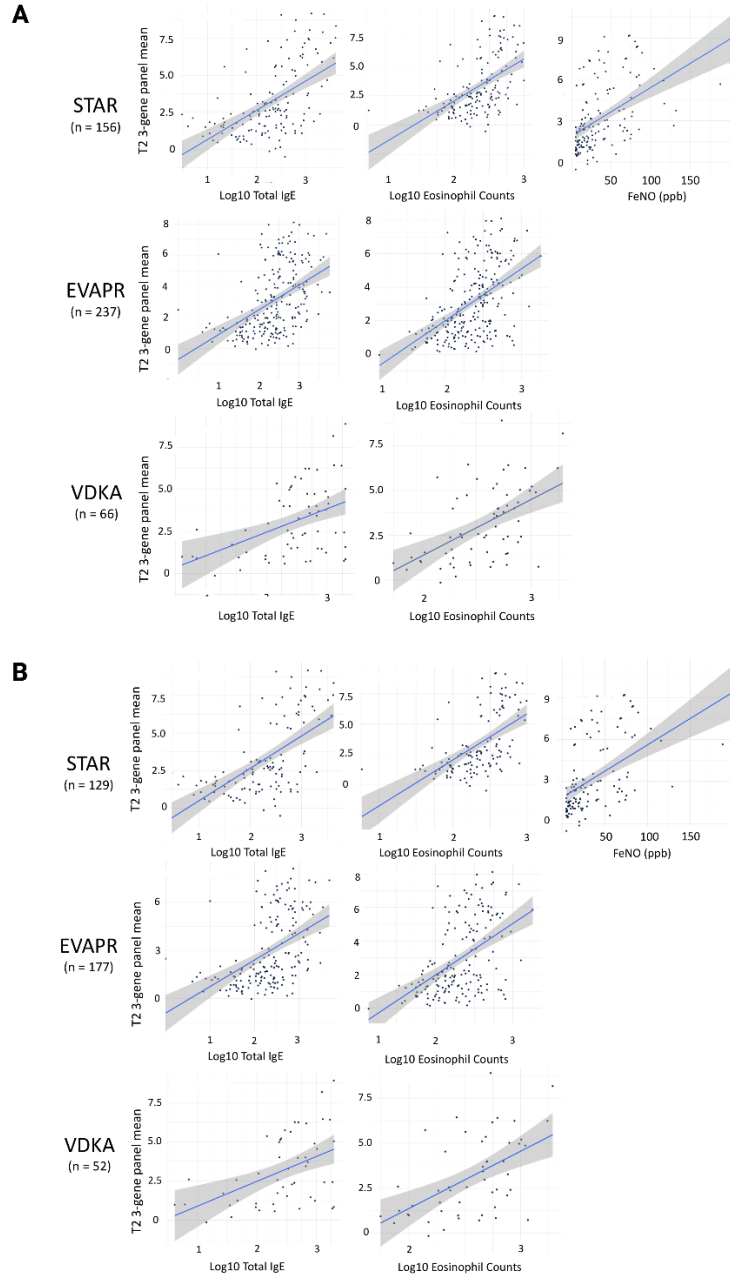


Legend:

Abbreviations: STAR, Stress and Treatment Response in Puerto Rican and African American Children with Asthma Study; EVA-PR, Epigenetic Variation and Childhood Asthma in Puerto Ricans Study, VDKA, Vitamin D Kids Asthma Study; PC, principal component.

Plots demonstrate that there is no clear clustering by cell sorting protocol in any study.

eFigure 5. Scatterplots demonstrating correlation between the T2 3-gene panel mean and log10 total IgE, log10 eosinophil count, and FeNO in the STAR, EVA-PR, and VDKA studies using K-means clustering profiles (Top Panel) and consensus cluster profiles (Bottom Panel)

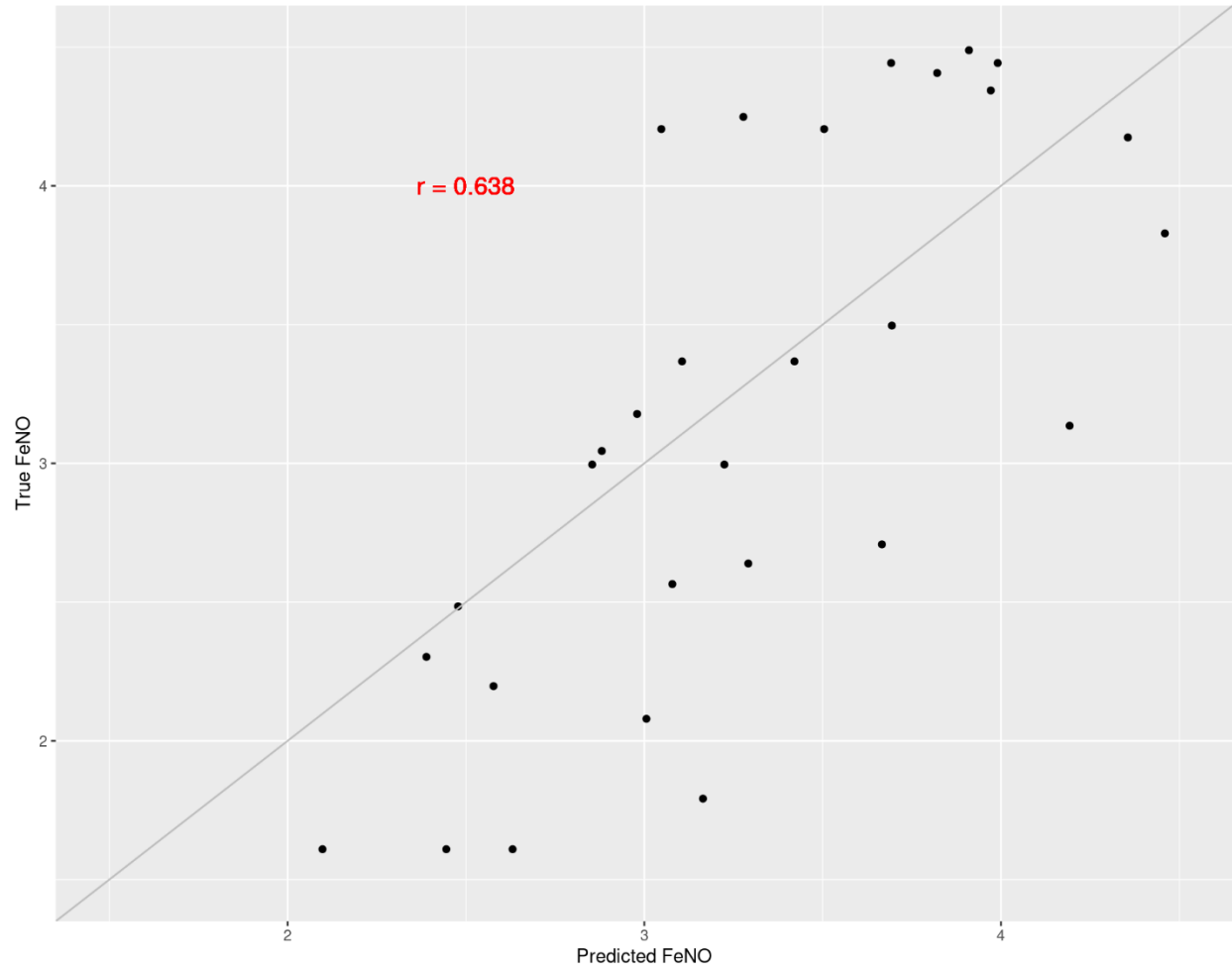


Legend:

Abbreviation: FeNO, Fractional exhaled nitric oxide; STAR, Stress and Treatment Response in Puerto Rican and African American Children with Asthma Study; EVA-PR, Epigenetic Variation and Childhood Asthma in Puerto Ricans Study; VDKA, Vitamin D Kids Asthma Study. FeNO level was only measured in STAR.

Total IgE and eosinophils (in all studies) and FeNO (in STAR) were moderately correlated with the T2 3-gene mean or 3GM ($\rho=0.43-0.59$, $p < 0.01$ in all instances).

eFigure 6. Prediction accuracy of predicted FeNO for the testing dataset in the STAR study

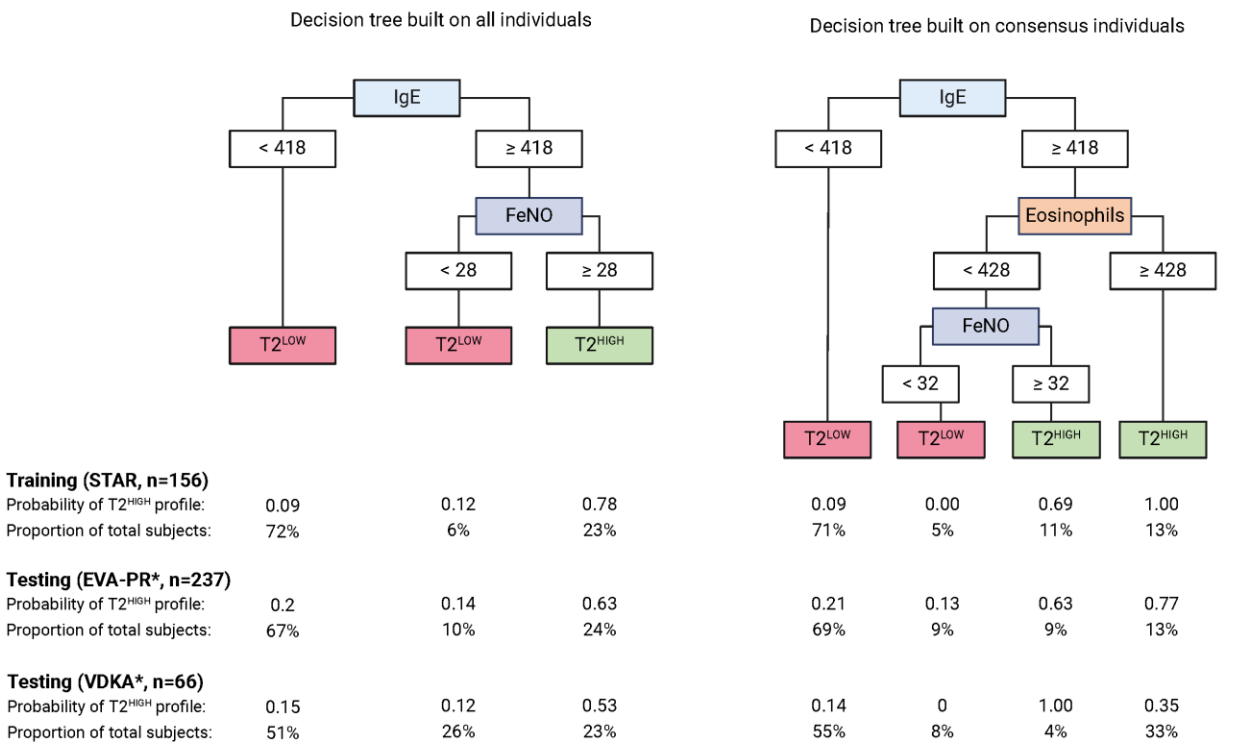


Legend:

Abbreviations: FeNO, fractional exhaled nitric oxide; STAR, Stress and Treatment Response in Puerto Rican and African American Children with Asthma Study; r, Pearson correlation coefficient.

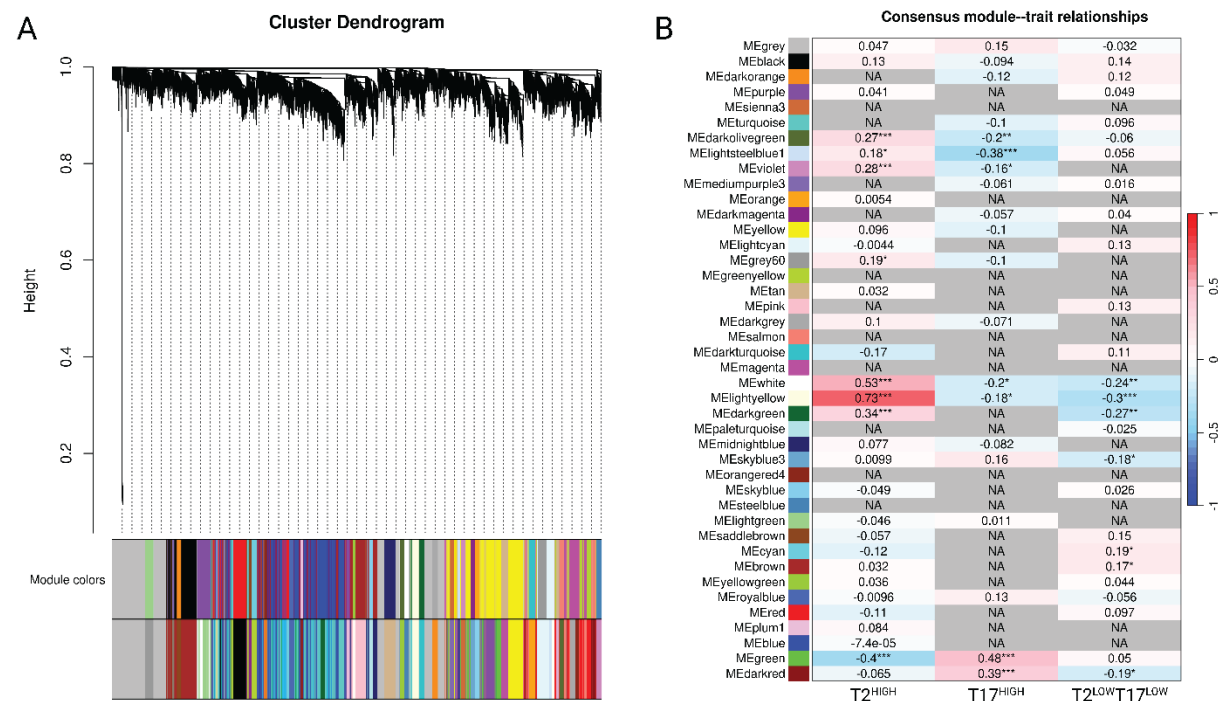
Predicted FeNO and true FeNO are shown in log scale on the x and y axes, respectively.

eFigure 7. Proposed decision tree for classification of T2^{HIGH} versus T2^{LOW} asthma and its performance in each study



Legend:
 Abbreviations: STAR, Stress and Treatment Response in Puerto Rican and African American Children with Asthma Study; EVA-PR, Epigenetic Variation and Childhood Asthma in Puerto Ricans Study; VDKA, Vitamin D Kids Asthma Study.
 Decimals represent the predicted probability of having a T2^{HIGH} profile for that set of conditions and percentages represent the percent of observations in that node for all individuals in each study. For the decision tree built on all samples, transcriptomic profile was determined by k-means clustering. For the decision tree built on consensus samples, transcriptomic profile was the same across all 5 clustering methods. Figure was created using BioRender.
 *indicates use of predicted (rather than measured) FeNO.

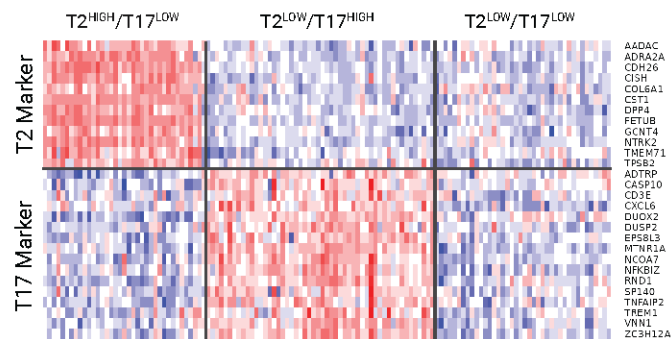
eFigure 8. Weighted gene co-expression network analysis cluster dendrogram and consensus-module trait relationships



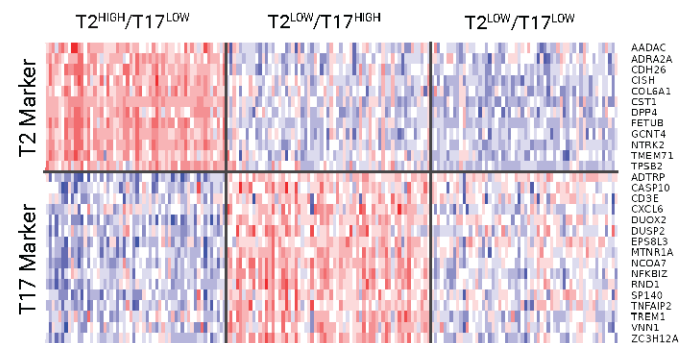
Legend:
Panel A: Cluster dendrogram of the co-expression modules identified by consensus weighted gene co-expression network analysis. The color row underneath the dendrogram shows the module assignment. Panel B: Heatmap illustrating the consensus module-trait relationships. Each cell contains the corresponding correlation and significance, with the color-coded by correlation according to the color legend.

eFigure 9. Heat maps of newly identified T2 and T17 genes in STAR, EVA-PR, and VDKA

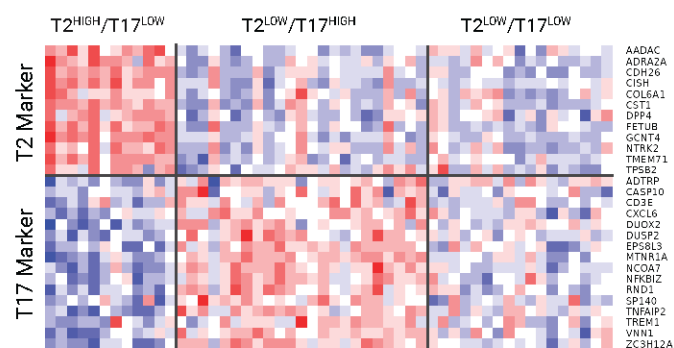
A: STAR (n=129)



B: EVA-PR (n=177)



C: VDKA (n=52)



Legend:

Abbreviations: STAR, Stress and Treatment Response in Puerto Rican and African American Children with Asthma Study; EVA-PR, Epigenetic Variation and Childhood Asthma in Puerto Ricans Study; VDKA, Vitamin D Kids Asthma Study.

Figure was created using BioRender.

eTable 1. Comparison of gene expression levels and clustering approaches by k-means cluster transcriptomic profile in STAR, EVA-PR, and VDKA

	Stress and Treatment Response in Puerto Rican and African American Children with Asthma Study n = 156			Epigenetic Variation and Childhood Asthma in Puerto Ricans Study n = 237			Vitamin D Kids Asthma Study n = 66		
Profile assignment by k-means clustering	T2 ^{HIGH} n = 40	T17 ^{HIGH} n = 57	T2 ^{LOW} / T17 ^{LOW} n = 59	T2 ^{HIGH} n = 69	T17 ^{HIGH} n = 83	T2 ^{LOW} / T17 ^{LOW} n = 85	T2 ^{HIGH} n = 15	T17 ^{HIGH} n = 31	T2 ^{LOW} / T17 ^{LOW} n = 20
T2 3-gene mean, mean (SD) ^a	6.72 (1.31)	1.86 (0.94)	2.00 (1.23)	5.53 (1.22)	2.44 (1.75)	1.75 (1.06)	5.85 (1.34)	2.27 (1.60)	2.23 (1.17)
T17 5-gene mean, mean (SD) ^b	4.02 (1.00)	5.61 (0.82)	3.07 (0.95)	3.25 (1.00)	5.50 (0.80)	3.16 (0.77)	3.14 (1.03)	6.32 (0.86)	4.00 (0.76)
Cluster assignment the same in k-means clustering and:									
Hierarchical clustering, No. (%)	36 (90.0)	53 (93.0)	48 (81.4)	66 (95.7)	75 (90.4)	61 (71.8)	14 (93.3)	28 (90.3)	15 (75.0)
K-medoid clustering, No. (%)	39 (97.5)	57 (100)	44 (74.6)	68 (98.6)	82 (98.8)	67 (78.8)	14 (93.3)	30 (96.8)	20 (100)
Gaussian mixture models clustering, No. (%)	40 (100)	53 (93.0)	42 (71.2)	57 (82.6)	71 (85.5)	81 (95.3)	12 (80.0)	31 (100)	20 (100)
Spectral clustering, No. (%)	40 (100)	56 (98.2)	49 (83.1)	68 (98.6)	70 (84.3)	75 (88.2)	15 (100)	26 (83.9)	20 (100)
All five clustering approaches, No. (%)	36 (90.0)	53 (93.0)	40 (67.8)	56 (81.2)	63 (75.9)	58 (68.2)	12 (80.0)	25 (80.6)	15 (75.0)
^a P-values for the comparison of the T2 3-gene in the T2 ^{HIGH} profile to the T17 ^{HIGH} profile and the T2 ^{LOW} /T17 ^{LOW} profile was <0.001 in all three studies.									
^b P-values for the comparison of the T17 5-gene in the T17 ^{HIGH} profile to the T2 ^{HIGH} profile and the T2 ^{LOW} /T17 ^{LOW} profile was <0.001 in all three studies.									

eTable 2. Adjusted Rand Index comparing five clustering approaches in STAR, EVA-PR, and VDKA

STAR	K means	Hierarchical	K medoids	Gaussian Mixture	Spectral
K means	1.000	0.667	0.725	0.631	0.792
Hierarchical	0.667	1.000	0.556	0.769	0.763
K medoids	0.725	0.556	1.000	0.708	0.709
Gaussian Mixture	0.631	0.769	0.708	1.000	0.814
Spectral	0.792	0.763	0.709	0.814	1.000
EVA-PR	K means	Hierarchical	K medoids	Gaussian Mixture	Spectral
K means	1.000	0.608	0.768	0.691	0.716
Hierarchical	0.608	1.000	0.679	0.500	0.619
K medoids	0.768	0.679	1.000	0.546	0.708
Gaussian Mixture	0.691	0.500	0.546	1.000	0.646
Spectral	0.716	0.619	0.708	0.646	1.000
VDKA	K means	Hierarchical	K medoids	Gaussian Mixture	Spectral
K means	1.000	0.677	0.851	0.892	0.862
Hierarchical	0.677	1.000	0.589	0.630	0.630
K medoids	0.851	0.589	1.000	0.835	0.766
Gaussian Mixture	0.892	0.630	0.835	1.000	0.760
Spectral	0.862	0.630	0.766	0.760	1.000

Abbreviations: STAR, Stress and Asthma Treatment Response in Puerto Rican and African American Children with Asthma Study; EVA-PR, Epigenetic Variation and Childhood Asthma in Puerto Ricans Study; VDKA, Vitamin D Kids Asthma Study.

The Adjusted Rand Index measures the similarity between two clustering approaches, ranging from 0 (no agreement) to 1 (perfect agreement). Higher values indicate greater similarity between the cluster results.

eTable 3. Average Proportion of Non-overlap for all participants ("All") and for participants with a consensus profile ("Consensus") in STAR, EVA-PR, and VDKA

	STAR		EVA-PR		VDKA	
	All	Consensus	All	Consensus	All	Consensus
Average proportion of non-overlap	0.1183	0.0476	0.2843	0.0302	0.1325	0.0457

Abbreviations: STAR, Stress and Asthma Treatment Response in Puerto Rican and African American Children with Asthma Study; EVA-PR, Epigenetic Variation and Childhood Asthma in Puerto Ricans Study; VDKA, Vitamin D Kids Asthma Study.

The average proportion of non-overlap shows the proportion of samples that are not placed in the same cluster when comparing the results from clustering based on the full data to clustering based on removing each column, one at a time.

eTable 4. Demographic and clinical characteristics, biomarkers, and lung function of participants in each study, by transcriptomic profile determined by k-means clustering

	Stress and Asthma Treatment Response in Puerto Rican and African American Children with Asthma Study (STAR) n = 156			Epigenetic Variation and Childhood Asthma in Puerto Ricans Study (EVA-PR) n = 237			Vitamin D Kids Asthma Study (VDKA) n = 66		
Variable	T2 ^{HIGH} n = 40	T17 ^{HIGH} n = 57	T2 ^{LOW} / T17 ^{LOW} n = 59	T2 ^{HIGH} n = 69	T17 ^{HIGH} n = 83	T2 ^{LOW} / T17 ^{LOW} n = 85	T2 ^{HIGH} n = 15	T17 ^{HIGH} n = 31	T2 ^{LOW} / T17 ^{LOW} n = 20
Age, mean (SD), years	14.2 (2.9)	13.2 (2.8)	15.1 (3.1)*	15.3 (3.0)	15.5 (2.7)	15.4 (2.8)	10.8 (2.6)	9.2 (2.5)	11.7 (2.3)*
Sex, No. (%)									
Male	19 (47.5)	32 (56.1)	22 (37.3)	37 (53.6)	52 (62.7)	46 (54.1)	8 (53.3)	21 (67.7)	10 (50.0)
Female	21 (52.5)	25 (43.9)	37 (62.7)	32 (46.4)	31 (37.3)	39 (45.9)	7 (46.7)	10 (32.3)	10 (50.0)
Ethnicity or race ^a , No. (%)									
African American or Black	17 (42.5)	45 (78.9)	50 (84.7)*	0 (0)	0 (0)	0 (0)	10 (66.7)	18 (58.1)	10 (50.0)
Other	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	4 (26.7)	3 (9.7)	3 (15.0)
Puerto Rican	23 (57.5)	12 (21.1)	9 (15.3)	69 (100)	83 (100)	85 (100)	0 (0)	0 (0)	0 (0)
White	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	1 (6.7)	10 (32.3)	7 (35.0)
Annual household income $\geq$ \$15,000 ^b ,	22/35 (62.9)	15/47 (31.9)	20/46 (43.5)*	29/69 (42.0)	32/83 (38.6)	37/84 (44.0)	7/13 (53.8)	20/31 (64.4)	10/18 (55.5)
No./total No. (%)									

≥1 parent completed some college education, No. (%)	32 (80.0)	33 (57.9)	36 (61.0)	36 (52.2)	46 (55.4)	49 (57.7)	10 (66.7)	22 (71.0)	12 (60.0)
Private or employer-based health insurance, No./total No. (%)	7/39 (17.9)	8/53 (15.1)	10/54 (18.5)	29/69 (42.0)	28/83 (33.7)	32/85 (37.6)	1/15 (6.7)	13/30 (41.9)	8/20 (40.0)*
Season at nasal and serum sampling visit, No./total No. (%)									
Sep-Nov	17/40 (42.5)	19/57 (33.3)	16/59 (27.1)	17/68 (25.0)	23/83 (27.7)	19/85 (22.4)	2/15 (13.3)	7/31 (22.6)	7/20 (35.0)*
Dec-Feb	11/40 (27.5)	9/57 (15.8)	15/59 (25.4)	17/68 (25.0)	21/38 (25.3)	16/85 (18.8)	7/15 (46.7)	6/31 (19.4)	6/20 (30.0)
Mar-May	6/40 (15.0)	16/57 (28.1)	16/59 (27.1)	18/68 (26.5)	21/83 (25.3)	25/85 (29.4)	1/15 (6.7)	17/31 (54.8)	5/20 (25.0)
June-Aug	6/40 (15.0)	13/57 (22.8)	12/59 (20.3)	16/68 (23.5)	18/83 (21.7)	25/85 (29.4)	5/15 (33.3)	1/31 (3.2)	2/20 (10.0)
Naso-ocular symptoms apart from colds ^c , past year, No./total No. (%)	29/38 (76.3)	31/51 (60.8)	40/55 (72.7)	64/69 (92.8)	69/83 (83.1)	70/85 (82.4)	10/14 (71.4)	22/29 (75.9)	13/20 (65.0)
Physician-diagnosed allergic rhinitis ^d , No./total No. (%)	16/38 (42.1)	6/50 (12.0)	17/54 (31.5)*	14/67 (20.9)	12/69 (14.5)	14/82 (17.1)	3/10 (30.0)	8/22 (36.4)	6/13 (46.2)
≥1 asthma-related ED/urgent care visit in the prior year, No./total No. (%)	26/39 (66.7)	23/49 (46.9)	16/54 (29.6)*	24/69 (34.8)	33/83 (39.8)	32/85 (37.7)	15/15 (100)	27/31 (87.1)	16/20 (80.0)
≥1 asthma-related hospitalization in the prior year, No. (%)	4/39 (10.3)	7/51 (13.7)	8/53 (15.1)	9/68 (13.2)	11/83 (13.3)	5/85 (5.9)	6/11 (54.5)	12/17 (70.6)	7/10 (70.0)
Body mass index ^e z-score, mean (SD)	0.96 (1.22)	1.38 (1.01)	1.18 (1.28)	0.74 (1.23)	0.88 (1.07)	0.88 (1.07)	1.22 (1.19)	0.62 (1.18)	1.34 (1.00)
Overweight or obesity ^f , No. (%)	21 (52.5)	43 (75.4)	38 (64.4)	28 (40.6)	40 (48.2)	40 (47.1)	12 (80.0)	12 (38.7)	13 (65.0)*

Use of nasal corticosteroids <4 weeks prior to nasal specimen collection, No. (%)	0 (0)	0 (0)	0 (0)	3 (4.4)	3 (3.6)	3 (3.5)	1 (6.7)	0 (0)	1 (5.0)
Use of inhaled corticosteroids <4 weeks prior to nasal specimen collection, No. (%)	0 (0)	0 (0)	0 (0)	19 (27.5)	21 (25.3)	22 (25.9)	15 (100)	31 (100)	20 (100)
ICS response ^g , No./total No.	15/39 (38.5)	18/52 (34.6)	23/55 (41.8)	-	-	-	-	-	-
≥1 IgE to allergens ≥0.35 IU/ml, No./total No. (%)	34/37 (91.9)	27/54 (50.0)	28/54 (51.9)*	64/67 (95.5)	53/78 (67.9)	48/84 (57.1)*	10/15 (66.7)	22/30 (73.3)	11/20 (55.0)
Serum total IgE, median (IQR), IU/ml	869 (421-1628) [n=37]	105 (38-311) [n=54]	183 (73-344) [n=54]*	584 (259-1096) [n=67]	216 (98-409) [n=78]	176 (66-414) [n=84]*	619 (279-1156)	382 (120-783) [n=30]	267 (109-484)
Blood eosinophil count, median (IQR), cells/ml	444 (287-559)	179 (114-297) [n=56]	164 (94-300) [n=57]*	343 (223-482)	187 (125-338) [n=82]	182 (102-322) *	560 (360-821)	413 (187-583) [n=30]	338 (193-507)
FeNO, median (IQR), ppb	59 (34.2-81.5) [n=38]	19 (10 - 34)	19 (8-33)*	-	-	-	-	-	-
FEV ₁ , mean (SD), %predicted	90.7 (15.7)	90.0 (17.0) [n=56]	92.8 (14.0) [n=58]	96.0 (16.1) [n=68]	97.4 (13.2) [n=80]	96.4 (15.9) [n=83]	90.7 (15.9)	90.0 (15.9)	94.6 (15.4)
FVC, mean (SD), %predicted	100.6 (14.1)	99.1 (13.5) [n=56]	101.7 (12.3) [n=58]	102.5 (14.5) [n=68]	102.2 (13.0) [n=80]	102.4 (14.7) [n=83]	105.0 (13.8)	99.7 (14.1)	103.7 (13.9)
FEV ₁ /FVC, mean (SD)	0.80 (0.07)	0.79 (0.11) [n=56]	0.81 (0.09) [n=58]	0.82 (0.08) [n=68]	0.84 (0.08) [n=80]	0.83 (0.07) [n=83]	0.76 (0.06)	0.80 (0.10)	0.80 (0.08)
Abbreviations: ED, emergency department; ICS, inhaled corticosteroid.									

* indicates $p < 0.05$ for the ANOVA (continuous variables, means), Kruskal-Wallis (continuous variables, medians), or chi-squared (categorical variables) test comparing $T2^{HIGH}$, $T17^{HIGH}$, and $T2^{LOW}/T17^{LOW}$ profiles in each study.

^aIn STAR, all participants had to have ≥ 3 Puerto Rican grandparents (at the Puerto Rico site) or ≥ 3 African American grandparents (at the Pittsburgh site). All EVA-PR participants had to have 4 Puerto Rican grandparents. In VDKA, parents selected one race option from the following categories: American Indian or Alaskan Native (AI/AN), Asian, Black or African American, Multiracial, Other, or White. In this table, small numbers were reported as the composite Other (which includes AI/AN, Asian, Multiracial, and Other) to protect participant anonymity.

^bAnnual household income $\geq \$30,000$ shown for VDKA and Pittsburgh participants in STAR.

^cNaso-ocular symptoms apart from colds was defined as a "Yes" response (by subject's parent) to each of the following questions: 1) Has the child had a runny or stuffy nose accompanied by sneezing and itching when he/she did not have a cold or flu?, 2) (If Yes to 1) Has this happened in the last 12 months?, and 3) Has the child ever had a runny or stuffy nose during the last 12 months as a result of exposure to any of the following: Pollens (grass, trees, weed, etc.), House Dust, Cats, or Dogs?

^dPhysician-diagnosed allergic rhinitis was ascertained by parental questionnaire.

^eThe body-mass index is the weight in kilograms divided by the square of the height in meters.

^fA body-mass index z-score $\geq 85^{th}$ percentile, calculated using growth charts from the U.S. Centers for Disease Control.

^gICS (inhaled corticosteroid) response was defined as an increment in $FEV_1 \geq 5\%$ from baseline after six weeks of ICS treatment.

eTable 5. Demographic and clinical characteristics, biomarkers, and lung function of participants in each study who were consistently in the same transcriptomic profile using five clustering approaches

Variable	Stress and Asthma Treatment Response in Puerto Rican and African American Children with Asthma Study (STAR) n = 129			Epigenetic Variation and Childhood Asthma in Puerto Ricans Study (EVA-PR) n = 177			Vitamin D Kids Asthma Study (VDKA) n = 52		
	T2 ^{HIGH}	T17 ^{HIGH}	T2 ^{LOW} / T17 ^{LOW}	T2 ^{HIGH}	T17 ^{HIGH}	T2 ^{LOW} / T17 ^{LOW}	T2 ^{HIGH}	T17 ^{HIGH}	T2 ^{LOW} / T17 ^{LOW}
	n = 36	n = 53	n = 40	n = 56	n = 63	n = 58	n = 12	n = 25	n = 15
Age, mean (SD), years	14.1 (2.9)	13.3 (2.7)	15.8 (3.0)*	15.6 (3.1)	15.8 (2.7)	15.4 (2.8)	10.8 (2.5)	8.8 (2.2)	11.6 (2.5)*
Sex, No. (%)									
Male	17 (47.2)	28 (52.8)	16 (40.0)	29 (51.8)	41 (65.1)	31 (53.4)	6 (50.0)	18 (72.0)	8 (53.3)
Female	19 (52.8)	25 (47.2)	24 (60.0)	27 (48.2)	22 (34.9)	27 (46.6)	6 (50.0)	7 (28.0)	7 (46.7)
Ethnicity or race ^a , No. (%)									
African American or Black	16 (44.4)	41 (77.4)	33 (82.5)*	0 (0)	0 (0)	0 (0)	9 (75.0)	14 (56.0)	8 (53.3)
Other	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	2 (16.7)	3 (12.0)	2 (13.3)
Puerto Rican	20 (55.6)	12 (22.6)	7 (17.5)	56 (100)	63 (100)	58 (100)	0 (0)	0 (0)	0 (0)
White	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	1 (8.3)	8 (32.0)	5 (33.3)
Annual household income $\geq$ \$15,000 ^b ,	20/31 (64.5)	12/43 (27.9)	15/31 (48.4)*	24 (42.9)	23 (36.6)	27/57 (47.4)	6/11 (54.5)	19 (76.0)	8 (53.3)
No./total No. (%)									

≥1 parent completed some college education, No. (%)	28 (77.8)	30 (56.6)	26 (65.0)	28 (50.0)	35 (55.6)	35 (60.3)	9 (75.0)	21 (84.0)	7 (46.6)*
Private or employer-based health insurance, No. (%)	5/35 (14.3)	8/49 (16.3)	9/36 (25.0)	24 (42.9)	20 (31.7)	24 (41.4)	1 (8.3)	12/24 (50.0)	4 (26.7)*
Season at nasal and serum sampling visit, No./total No. (%)									
Sept-Nov	16 (44.4)	17 (32.1)	13 (32.5)	13 (23.6)	22 (34.9)	12 (20.7)	2 (16.7)	5 (20.0)	5 (33.3)*
Dec- Feb	10 (27.8)	8 (15.1)	8 (20.0)	16 (29.1)	17 (27.0)	9 (15.5)	4 (33.3)	4 (16.0)	4 (26.7)
Mar- May	5 (13.9)	16 (30.2)	11 (27.5)	14 (25.5)	11 (17.5)	16 (27.6)	1 (8.3)	15 (60.0)	5 (33.3)
June- Aug	5 (13.9)	12 (22.6)	8 (20.0)	12 (21.8)	13 (20.6)	21 (36.2)	5 (41.7)	1 (4.0)	1 (6.7)
Naso-ocular symptoms apart from colds ^c , past year, No./total No. (%)	26/32 (76.5)	27/47 (57.4)	28/38 (73.7)	53 (94.6)	51 (81.0)	46 (79.3)*	8/11 (72.7)	18/24 (75.0)	8 (53.3)
Physician-diagnosed allergic rhinitis ^d , No./total No. (%)	14/34 (41.2)	5/46 (10.9)	11/38 (28.9)*	13/54 (24.1)	9 (14.3)	10/55 (18.2)	3/8 (37.5)	5/18 (27.8)	4/8 (50.0)
≥1 asthma-related ED/urgent Care visit in the prior year, No./total No. (%)	23/35 (65.7)	19/45 (42.2)	11/36 (30.6)*	20 (35.7)	26 (41.3)	22 (37.9)	12 (100)	23/25 (92)	13/15 (86.7)
≥1 asthma-related hospitalization in the prior year, No./total No. (%)	4/35 (11.4)	7/47 (14.9)	5/36 (13.9)	8/55 (14.5)	10/63 (15.9)	3 (5.2)	6/9 (66.7)	10/15 (66.7)	6/8 (75.0)
Use of nasal corticosteroids <4 weeks prior to nasal specimen collection, No. (%)	0 (0)	0 (0)	0 (0)	2 (3.6)	2 (3.2)	3 (5.2)	1 (8.3)	0 (0)	1 (6.7)

Use of inhaled corticosteroids <4 weeks prior to nasal specimen collection, No. (%)	0 (0)	0 (0)	0 (0)	16 (28.6)	13 (20.6)	17 (29.3)	12 (100)	25 (100)	15 (100)
Body mass index ^e z-score, mean (SD)	1.01 (1.21)	1.34 (1.02)	1.21 (1.35)	0.71 (1.16)	0.84 (1.02)	0.89 (1.04)	1.11 (1.29)	0.75 (1.14)	1.30 (1.04)
Overweight or obesity ^f , No. (%)	19 (52.8)	40 (75.5)	27 (67.5)	22 (39.3)	29 (46.0)	28 (48.3)	9 (75.0)	10 (40.0)	10 (66.7)
ICS response ^g , No./total No. (%)	14/35 (40.0)	18/48 (37.5)	15/39 (40.5)	-	-	-	-	-	-
≥1 IgE to allergens ≥0.35 IU/ml, No./total No. (%)	31/33 (93.9)	24/50 (48.0)	19/37 (51.4)*	53/55 (96.4)	41/62 (66.1)	29 (50.0)*	7 (58.3)	18/24 (75.0)	7 (46.7)
Serum total IgE, median (IQR), IU/ml	881 (460-1628)	83 (33.5-275.5)	183 (104-372)	584 (288-1013)	206.5 (59.5-344)	141.5 (51.2-392.5)*	594 (289-1272.5)	347 (96.8-844)	223 (72-430.5)
	[n=33]	[n=50]	[n=37]*	[n=55]	[n=62]			[n=24]	
Blood eosinophil count, median (IQR), cells/ml	455.1 (321.3-591.9)	171.7 (99.5-280.5) [n=52]	165 (84.3-283.5) [n=38]*	344.9 (233.2-526.2)	164.7 (111.7-278) [n=62]	134.3 (79.9-288)*	550 (406-915)	456.5 (184.5-625.5) [n=24]	235 (187-467.5)
FeNO, median (IQR), ppb	59 (34.2-81.5) [n=34]	16 (10-29)	16 (6.8-41.2)*	-	-	-	-	-	-
FEV ₁ , mean (SD), %predicted	90.8 (15.8)	89.9 (17.8) [n=52]	92.2 (12.8)	95.3 (16.1) [n=55]	98.1 (13.5) [n=61]	96.2 (16.8) [n=56]	88.8 (13.5)	92.7 (14.4)	93.0 (13.4)
FVC, mean (SD), %predicted	100.2 (14.3)	99.4 (13.9) [n=52]	102.1 (11.1)	101.8 (15.0) [n=55]	103.5 (13.8) [n=61]	101.0 (15.5) [n=56]	103.9 (13.7)	101.2 (14.7)	100.5 (12.8)
FEV ₁ /FVC, mean (SD)	0.80 (0.08)	0.79 (0.11) [n=52]	0.80 (0.10)	0.83 (0.08) [n=55]	0.84 (0.08) [n=61]	0.84 (0.07) [n=56]	0.75 (0.06)	0.81 (0.09)	0.81 (0.08)
Abbreviations: ED, emergency department; ICS, inhaled corticosteroid.									

* indicates $p < 0.05$ for the ANOVA (continuous variables, means), Kruskal-Wallis (continuous variables, medians), or chi-squared (categorical variables) test comparing $T2^{HIGH}$, $T17^{HIGH}$, and $T2^{LOW}/T17^{LOW}$ profiles in each study.

^aIn STAR, all participants had to have ≥ 3 Puerto Rican grandparents (at the Puerto Rico site) or ≥ 3 African American grandparents (at the Pittsburgh site). All EVA-PR participants had to have 4 Puerto Rican grandparents. In VDKA, parents selected one race option from the following categories: American Indian or Alaskan Native (AI/AN), Asian, Black or African American, Multiracial, Other, or White. In this table, small numbers were reported as the composite Other (which includes AI/AN, Asian, Multiracial, and Other) to protect participant anonymity.

^bAnnual household income $\geq \$30,000$ shown for VDKA and Pittsburgh participants in STAR.

^c Naso-ocular symptoms apart from colds was defined as a "Yes" response (by subject's parent) to each of the following questions: 1) Has the child had a runny or stuffy nose accompanied by sneezing and itching when he/she did not have a cold or flu?, 2) (If Yes to 1) Has this happened in the last 12 months?, and 3) Has the child ever had a runny or stuffy nose during the last 12 months as a result of exposure to any of the following: Pollens (grass, trees, weed, etc.), House Dust, Cats, or Dogs?

^dPhysician-diagnosed allergic rhinitis was ascertained by parental questionnaire.

^eThe body-mass index is the weight in kilograms divided by the square of the height in meters.

^fA body-mass index z-score $\geq 85^{th}$ percentile, calculated using growth charts from the U.S. Centers for Disease Control.

^gICS (inhaled corticosteroid) response was defined as an increment in $FEV_1 \geq 5\%$ from baseline after six weeks of ICS treatment.

^hCluster assignment determined by k-means clustering

eTable 6. Comparison of demographic and clinical characteristics for participants in STAR by ICS responsiveness within each of three transcriptomic profiles, defined by k-means clustering

Variable	T2 ^{HIGH}		T17 ^{HIGH}		T2 ^{LOW} /T17 ^{LOW}	
	n = 39		n = 52		n = 55	
	Responder n = 15	Non-Responder n = 24	Responder n = 18	Non-Responder n = 34	Responder n = 23	Non-Responder n = 32
Age, mean (SD), years	13.9 (3.0)	14.4 (2.8)	13.6 (3.3)	13.1 (2.7)	14.5 (3.0)	15.5 (3.2)
Sex, No. (%)						
Male	5 (33.3)	13 (54.2)	7 (38.9)	23 (67.6)*	7 (30.4)	13 (40.6)
Female	10 (66.7)	11 (45.8)	11 (61.1)	11 (32.4)*	16 (69.6)	19 (59.4)
Ethnicity or race ^a , No. (%)						
African American	4 (26.7)	12 (50.0)	2 (11.1)	8 (23.5)	3 (13.0)	4 (12.5)
Puerto Rican	11 (73.3)	12 (50.0)	16 (88.9)	26 (76.5)	20 (87.0)	28 (87.5)
Annual household income ≥\$15,000 ^b , No./total No. (%)	10/14 (71.4)	14/20 (70.0)	7/13 (53.8)	15/29 (51.7)	13/19 (68.4)	18/24 (75.0)
≥1 parent completed some college education, No. (%)	13 (86.7)	19 (79.2)	11 (61.1)	19 (55.9)	16 (69.6)	19 (59.4)
Private or employer-based health insurance, No./total No. (%)	11 (73.3)	12/23 (52.2)	4/17 (23.5)	10/31 (32.3)	6/21 (28.6)	7/29 (24.1)
≥1 asthma-related ED/urgent care visit in the prior year, No./total No. (%)	8 (53.3)	17/23 (73.9)	6/15 (40.0)	14/29 (48.3)	7/20 (35.0)	7/31 (22.6)

≥1 asthma-related hospitalization in the prior year, No./total No. (%)	1 (6.7)	3/23 (13.0)	1/16 (6.2)	3/30 (10.0)	3/21 (14.3)	4/29 (13.8)
Body mass index ^c Z-score, mean (SD)	1.12 (1.13)	0.88 (1.31)	1.46 (1.04)	1.36 (1.06)	0.97 (1.38)	1.24 (1.24)
Overweight or obesity ^d , No. (%)	8 (53.3)	13 (54.2)	14 (77.8)	25 (73.5)	14 (60.9)	20 (62.5)
Th2 3-gene ^e mean, mean (SD)	7.02 (1.56)	6.56 (1.16)	1.66 (0.69)	2.02 (1.05)	1.87 (1.05)	1.99 (1.36)
Th17 5-gene ^f mean, mean (SD)	3.92 (0.94)	4.06 (1.07)	5.41 (0.51)	5.65 (0.95)	3.37 (0.70)	2.79 (1.07)
FeNO, median (IQR), ppb	77 (68-82) [n=13]	46 (34-74)	15 (14-30)	21 (8-34)	20 (12-38)	17 (7 - 31)
Change in FeNO, median (IQR), ppb	-35 (-48 - -11) [n=13]	-9 (-22.5 - 4.8)*	-3 (-8.5 - -1.5)	-1.5 (-12.8 - 2.0)	-1.0 (-16.0 - 0.5)	0 (-5.5 - 2.0)
Total IgE, median (IQR), IU/mL	1105 (750-1475)	629 (321-1787) [n=21]	66.5 (24-251.2)	113 (47-230) [n=33]	181.5 (47.2- 345.8) [n=22]	227.5 (135.8-344) [n=30]
Eosinophil count, median (IQR), cells/mL	423.4 (282.3- 658.4)	459.2 (287.2- 522.8)	126 (93-210) [n=17]	209.1 (145.7- 375.8)*	201 (106-318) [n=21]	151 (83.8-282.2)
FEV ₁ , mean (SD), %predicted	86.3 (11.9)	93.5 (17.6)	85.2 (19.5)	93.3 (16.0)	86.4 (13.2)	97.4 (13.1)*
FVC, mean (SD), %predicted	96.9 (10.9)	102.8 (15.8)	101.4 (10.8)	100.0 (14.2)	100.3 (12.0)	102.7 (13.0)
FEV ₁ /FVC ratio, mean (SD)	0.79 (0.09)	0.80 (0.07)	0.74 (0.14)	0.82 (0.07)*	0.77 (0.11)	0.84 (0.06)*
Abbreviations: STAR, Stress and Treatment Response in Puerto Rican and African American Children with Asthma Study; ICS, inhaled corticosteroid; ED, emergency department; ppb, parts per billion.						

Data is shown in the table for the participants in STAR who completed pulmonary function testing at both the enrollment visit and the follow up visit after 6 weeks of treatment with medium-dose inhaled Mometasone (n=146). Steroid response was not measured in EVA-PR or VDKA, so analysis was limited to STAR. ICS response was defined as $\geq 5\%$ increment in FEV₁ from baseline, after 6 weeks of ICS treatment.

Values shown in table represent *mean (standard deviation), median (Q1-Q3), or number (%)*.

* indicates $p < 0.05$ for the t-test (continuous, means), Wilcoxon (continuous, medians), or chi-squared (categorical) test comparing those with and those without an ICS response in each transcriptomic endotype.

^aFor STAR, "African American" was defined as having ≥ 3 African American grandparents, and "Puerto Rican" was defined as having ≥ 3 Puerto Rican grandparents.

^bAnnual household income $\geq \$30,000$ shown for VDKA and Pittsburgh participants in STAR.

^cThe body-mass index is the weight in kilograms divided by the square of the height in meters.

^dA body-mass index z-score $\geq 85^{\text{th}}$ percentile, calculated using growth charts from the U.S. Centers for Disease Control.

eTable 7. Predicted FeNO levels for participants in the EVA-PR and VDKA studies

Variable	EVA-PR			VDKA		
	n = 237			n = 48		
	T2 ^{HIGH}	T17 ^{HIGH}	T2 ^{LOW} / T17 ^{LOW}	T2 ^{HIGH}	T17 ^{HIGH}	T2 ^{LOW} / T17 ^{LOW}
	n = 69 (29.1)	n = 83 (35.0)	n = 85 (35.9)	n = 11 (22.9)	n = 25 (52.1)	n = 12 (25.0)
Predicted FeNO level (ppb)	44.7 [35.3 - 61.8]	26.3 [20.3 - 40.1]	23.2 [18.2 - 30.5]*	33.9 [31.9 - 40.7]	20.1 [15.4 - 30.6]	23.2 [19.6 - 26.9]*
Predicted FeNO level $\geq$ 20 ppb	69 (100.0)	62 (74.7)	56 (65.9)*	10 (90.9)	13 (52.0)	9 (75.0)
Predicted FeNO level $\geq$ 35 ppb	52 (75.4)	25 (30.1)	16 (18.8)*	4 (36.4)	5 (20.0)	2 (16.7)
Predicted FeNO level $\geq$ 50 ppb	29 (42.0)	11 (13.3)	4 (4.7)*	2 (18.2)	1 (4.0)	0 (0.0)
Abbreviations: EVA-PR, Epigenetic Variation and Childhood Asthma in Puerto Ricans Study; VDKA, Vitamin D Kids Asthma Study; FeNO, fractional exhaled nitric oxide.						
An elastic-net prediction model utilizing a set of 1,013 pre-selected genes was employed to predict FeNO.						

eTable 8. Random Forest model for prediction of T2^{HIGH} cluster profile in each of the three cohorts

Decision Tree	All Participants			Participants with a Consensus Profile		
	Train on STAR	Test on EVA-PR	Test on VDKA	Train on STAR	Test on EVA-PR	Test on VDKA
Accuracy	0.88	0.77	0.79	0.90	0.79	0.71
Area under the curve	0.83	0.67	0.69	0.88	0.73	0.69
Sensitivity	0.93	0.88	0.86	0.95	0.91	0.73
Specificity	0.73	0.51	0.53	0.78	0.52	0.67
Random Forest	All Participants			Participants with a Consensus Profile		
	10-Fold CV on STAR	Test on EVA-PR	Test on VDKA	10-Fold CV on STAR	Test on EVA-PR	Test on VDKA
Accuracy	0.85	0.77	0.66	0.85	0.77	0.67
Area under the curve	0.82	0.78	0.71	0.88	0.81	0.72
Sensitivity	0.93	0.90	0.74	0.93	0.93	0.69
Specificity	0.60	0.45	0.40	0.65	0.44	0.58

Abbreviations: CV, cross-fold validation; STAR, Stress and Treatment Response in Puerto Rican and African American Children with Asthma Study; EVA-PR, Epigenetic Variation and Childhood Asthma in Puerto Ricans Study; VDKA, Vitamin D Kids Asthma Study.

Red marks show better performance when comparing decision tree and random forest.

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